

Supplemental Information

Look Out Below: Predicting Wastewater Infrastructure Service Type at the Land Parcel Scale

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Section S1: Algorithm Selection

It is important to test several machine learning algorithms before selecting one for final use because a priori assumptions of what the best algorithm to use for a given problem may be incorrect. We tested four machine learning algorithms and evaluated their performance across three metrics: accuracy, F1-score, and ROC-AUC. We tested classification trees (CT), random forest (RF), single layer feed-forward neural networks (NN), and support vector machines (SVM) (ROC-AUC could not be calculated for SVM). Classification trees are tree-based models which are highly interpretable and able to handle many types of predictors (Kuhn and Johnson, 2013). Random forests combine tree predictors where trees within a forest depend on values of a random vector sampled independently and with the same distribution (Breiman, 2001). Random forests typically have higher accuracy compared to classification trees and overfitting is prevented. Random forests are less readily interpretable than classification trees. The NN method tested is the multilayer perceptron (MLP), which is based on a simple model of biological brains and often used for analyzing very large datasets. MLPs have low interpretability and higher computational cost compared to the tree-based methods tested. SVMs try to maximize the width of the margin between classes (using a linear class boundary). SVMs are known to be mathematically complex and computationally intensive (Suthaharan, 2016).

To evaluate which algorithm would work best for the classification problems here, we first selected 10 random samples (outer folds) of 100,000 parcels with known onsite and sewer labels from the FLWMI database. Equal amounts of onsite and sewer labels were used for each fold to ensure class balance. For each of these folds, 10-fold cross validation splits with 70/30 training/validation apportionment were built. For each of these cross-validation folds, a 10-parameter grid search was used to evaluate model performance under different hyperparameter sets (Table S1).

Table S1. Machine learning engines and hyperparameters tuned for each algorithm tested

Model	Engine	Hyperparameters Tuned	Citation
CT	rpart	Cost complexity, tree depth	(Therneau et al., 2015)
RF	ranger	mtry (# predictors randomly sampled at each split), min_n (minimum # data points in a node required for a split).	(Wright and Ziegler, 2017)
NN	nnet	# hidden units, penalty (amount of weight decay)	(Chollet and others, 2015)
SVM	Liblinear	Cost	(Helleputte et al., 2024)

Models were evaluated by their performance (accuracy, F1-score, and ROC-AUC) on the validation partition of their corresponding cross-validation fold. Additionally, model training time was considered. In the set up for developing these models in tidymodels, and all other models mentioned moving forward,

the ‘recipes’ package was used to process data for use in models. This includes setting certain variables specifically for identification purposes (model output retains these variables but does not train the model based on them), removing any training features with zero variance (i.e., the value of the feature is identical for all rows in the data partition), and data normalization (normalizing numeric data to have a standard deviation of one and a mean of zero). Using the recipes package prevents data leakage that could be imposed through removing features with zero variance and data normalization, since these processing steps are applied separately to the training and validation or test partitions. The seed for the random number generator was set for reproducibility at all necessary stages including developing outer folds, cross-validation folds, and RF ranger engine.

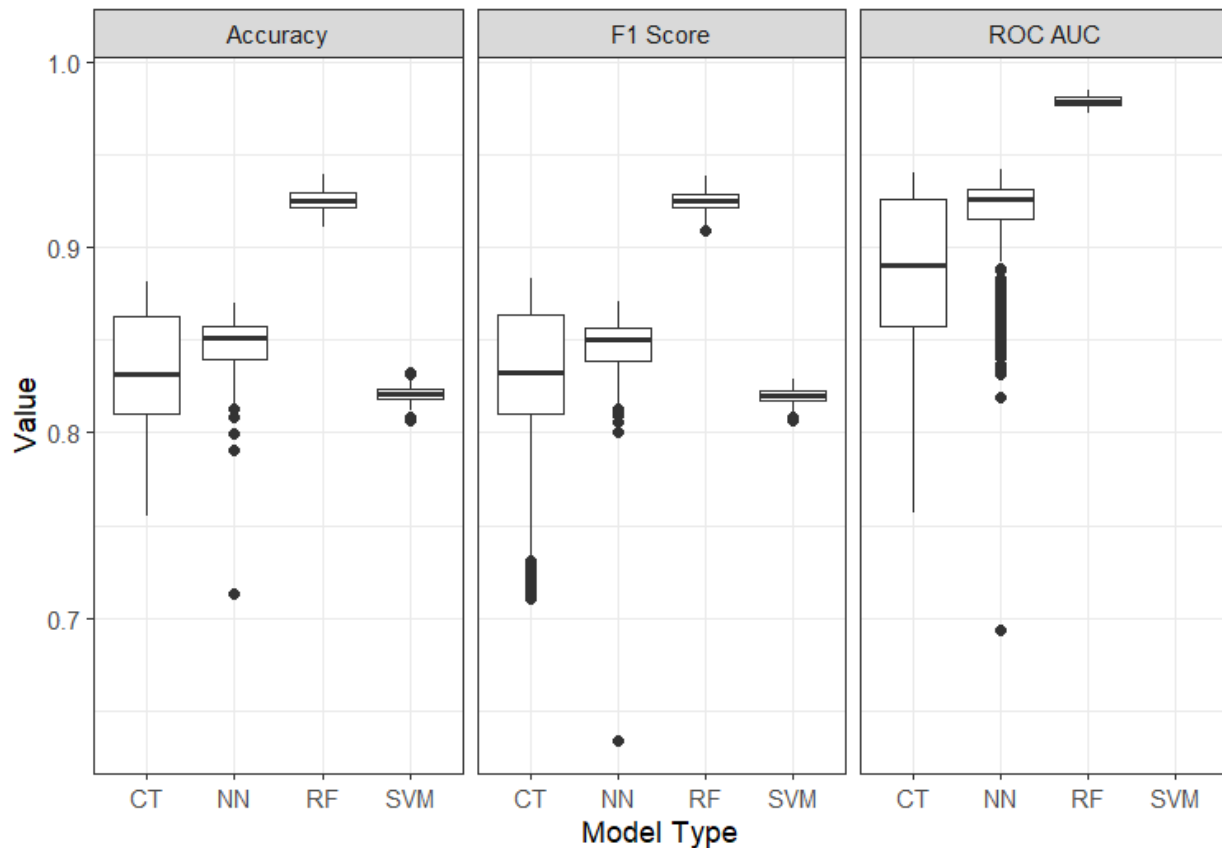


Figure S1: Boxplots of algorithm performance metrics for classification trees (CT), neural networks (NN), support vector machine (SVM), and random forests (RF). Each panel title indicates the performance metric being evaluated. Each boxplot represents the results for the metric shown across 10 outer sample folds of 100,000 rows with 10-fold cross validation on the training partition, with 10 potential hyperparameter sets chosen by grid-search.

The random forest algorithm had the highest performance across all 3 metrics of performance evaluated, F1-score, and ROC-AUC (Figure S1). RF also showed the least variation in performance between sample folds, cross-validation, and hyperparameter sets used to train the model. RF performed better than CT likely due to RF being dependent on the aggregation of many classification trees, and better than NN potentially because the tree-based RF can outperform NNs when tabular data (i.e. the type of data used here) are used (Grinsztajn et al., 2022). While CTs are more interpretable than RF and NN, there was a significant cost to performance.

Section S2: Evaluating model sensitivity to training data volume

In developing models, model performance and time required for training should be weighed against each other. To evaluate the benefits of adding additional training data, we selected 10 random samples (outer folds) of varying numbers of parcels (ranging from 50 thousand to 1 million) with known onsite and sewer labels from the FLWMI database. Equal amounts of onsite and sewer labels (50/50 split) were used for each fold to ensure class balance. For each fold, 10-fold cross validation splits with 70/30 training/validation apportionment were built. For each of these cross-validation folds, a 10-parameter grid search was used to evaluate model performance under different hyperparameter sets. The random forest algorithm was selected based on the results of the previous experiment on algorithm selection. Models were evaluated by their performance (accuracy, F1-score, and ROC-AUC) on the validation partition of their corresponding cross-validation fold. Additionally, the training time required to build the model was evaluated. 48 cores were used for parallel processing in building the random forest models. 256 trees were built for each random forest model (Oshiro et al., 2012). Selecting enough rows to balance against training time was done before moving on to feature selection experiments because adjusting the number of rows is simpler than adjusting the number and inclusion of features.

Performance metric values increased as training data volume increased (Fig. S2). The average training time increased when the number of rows used for training increased, peaking at ~173 minutes when 1 million rows were used (Fig. S3). The maximum model training time of only 173 minutes suggested that training data volume is not a limiting factor for the model because while training time will increase as data volume increases, the threshold for when it becomes computationally unreasonable to run the model likely corresponds to a much higher training data volume based on our experiments. Because the computational cost was not prohibitive, 1 million rows of training data was used for the next experiments.

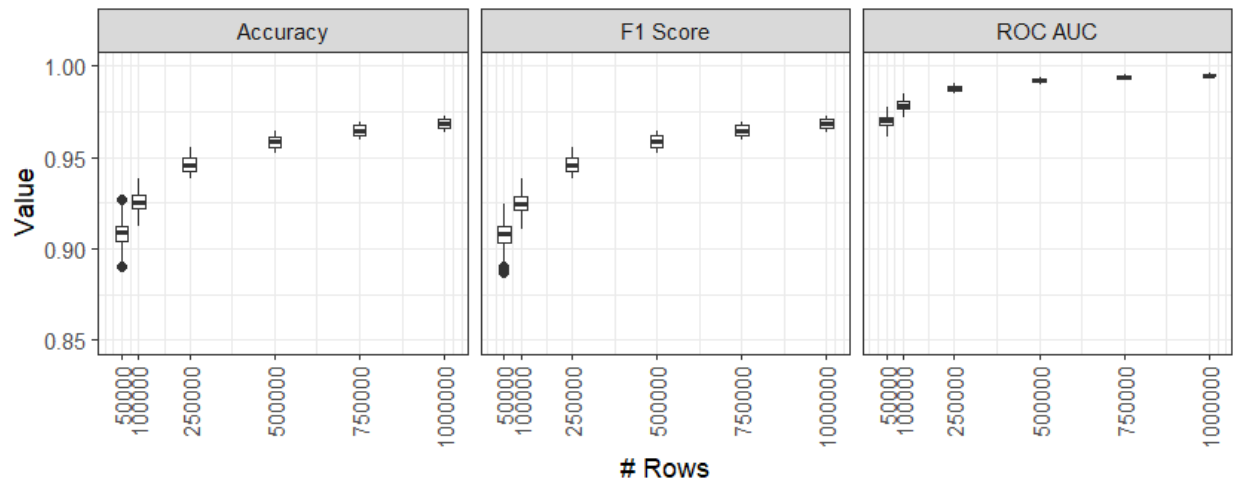


Figure S2. Boxplots of performance metrics when varying numbers of rows of data are used for training a random forest model. Each panel title indicates the performance metric being evaluated. Each boxplot represents the results for the metric shown across 10 outer sample folds of n rows with 10-fold cross validation on the training partition, with 10 potential hyperparameter sets chosen by grid-search.

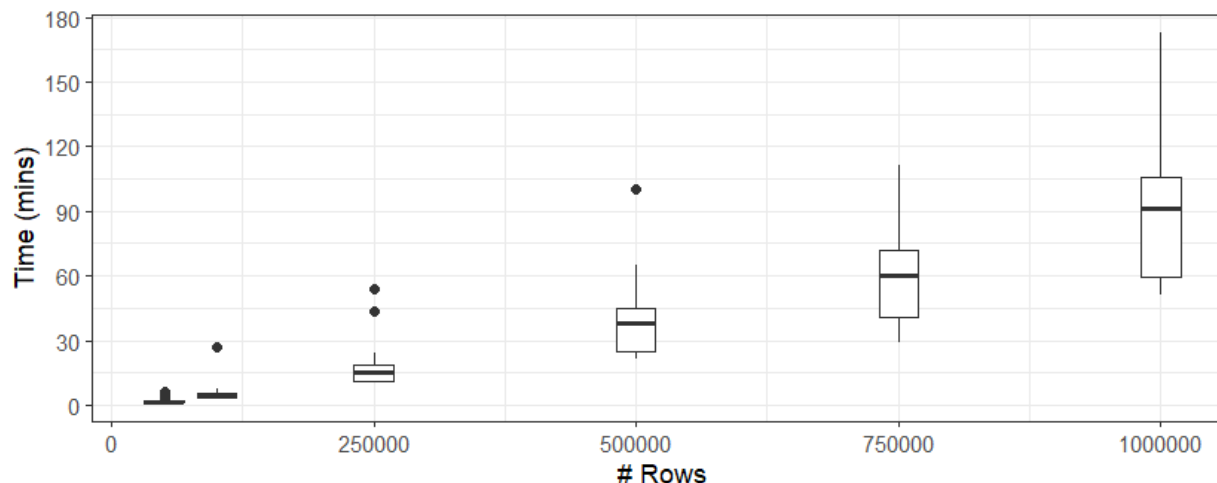


Figure S3. Model Training time versus number of rows sampled for training. Each boxplot represents the results for the metric shown across 10 outer sample folds of n rows with 10-fold cross validation on the training partition, with 10 potential hyperparameter sets chosen by grid-search.

Section S3. Feature Selection

The experiments demonstrated that the random forest algorithm and 1,000,000 rows of training data were sufficient to balance model performance against model training time. Feature selection experiments aid in identifying features to remove to reduce computational time while maintaining model performance. We ran three experiments: 1) removal of features across the training dataset based on near zero variance and degree of correlation, 2) maintaining or removing input features based on human logic and computational burdens, and 3) removing wastewater facility data due to gaps in feature availability and substituting ZTRAX data.

FS1: We started with identifying variables used that had near-zero variance; predictors with a single unique value for the entire dataset or that have few unique values relative to the number of samples (1:10 ratio unique/total) and where the ratio of the frequency of the most common value to the second most common value is large ($>95/5$). Similarly, variables with an absolute correlation value >0.9 are removed, the pre-processing step to accomplish this task (step_corr function in the recipes package) tries to remove the minimum number of variables so that all the resulting absolute correlations are less than 0.9 (or another user-specified threshold).

FS2: The remaining variables from FS1 were grouped by their respective dataset and computational process required to generate the values for those variables. These variables were compared to the original set of variables used to ensure that the process retained variables in a logical manner. For example, certain variables from the ACS dataset should be retained because they are part of a clear group such as varying number of rooms in housing units within a census block. Certain road related variables were marked for exclusion based on variable importance and computational requirements. The variable importance of features was evaluated and datasets with consistently low rankings were marked for exclusion. This set of features was also evaluated for performance with the Stage 1 classification.

FS3: Removing the permitted flow from wastewater treatment plants was evaluated because this information is somewhat sparsely populated in the wastewater treatment plant dataset used. The ZTRAX variables were removed due to future data availability concerns (Zillow's ZTRAX program has been

discontinued). In substitution, parcel value from Regrid's dataset was used in place of the tax amount for the parcel from ZTRAX.

The method of evaluation was to take the sample folds originally developed for training and testing the number of rows corresponding to 1,000,000 rows of training data then to run the model with the corresponding variables removed. For these experiments, each of the 10 folds of training data was tested against a corresponding test dataset; no cross-validation and no hyperparameter tuning were performed for this process as previous experiments demonstrated that hyperparameter tuning had limited effect on model performance outcomes. Accuracy across sample folds and different variable selections were evaluated.

Feature selection experiments showed similarly high resulting accuracies regardless of the formulation being tested (Table S2) with a range of 95.92-97.35% accuracy across all experiments. FS1-3 each had comparable accuracy to the initial feature set used. By using fewer features computational times required to build models is reduced and time to calculate the values for each of the features is also reduced for some of the input feature datasets. Ultimately, the FS3 formulation was chosen due to its high performance and that by excluding wastewater permitted flows and ZTRAX data, concerns with availability of data for that feature in states besides Florida and future model iterations could be avoided.

Table S2: Feature selection experiments. Number of variables retained in each input feature dataset in feature selection experiments is shown.

Roads	Buildings and Parcels	Landcover	Wastewater Facilities	Places	ZTRAX	ACS	Total	Accuracy Range %
<i>Feature Selection (FS) Experiment Formulation:</i>								
Initial Set of Features								
88	28	16	2	3	7	22	166	96.3-97.2
<i>Feature Selection (FS) Experiment Formulation:</i>								
FS1: Features with Near-Zero-Variance or a high degree of correlation (>0.9) removed								
10	20	4	2	3	4	13	56	95.9-97.1
<i>Feature Selection (FS) Experiment Formulation:</i>								
FS2: Changes compared to FS1.								
Roads: Only two specific road types retained, only maximum based features retained (means removed)								
Landcover: completely removed due to low variable importance and high computational time required for preprocessing								
ACS: Features added back in to ensure ACS variable groupings retained								
12	20	0	2	3	4	22	63	97.30-97.34
<i>Feature Selection (FS) Experiment Formulation:</i>								
FS3: Removing wastewater facility permitted flow feature and replacement of ZTRAX features with Regrid parcel value								
12	20	0	1	3	0 (1)*	22	59	96.5-97.3

*Parcel value from Regrid replaced tax amount from ZTRAX.

In FS1, where features with near zero variance and high correlation were removed, a smaller set of features adequately described each parcel. In FS2, considerations for variable inclusion and exclusion were based primarily on computational cost needed to generate the features and variable importance of the features. The computational effort required to generate features exceeded the effort required to build an associated RF model, so we considered logically consistent ways to reduce the generated features generated, including retaining non-redundant road-based features (maximum values associated with road

length and quantity in grid cells provided similar predictive capability to mean values) and informative ones (features based on less common road types had lower variable importance). When evaluating variable importance of FS1, landcover features consistently ranked lowest and were therefore removed for FS2. In FS3, permitted flow of the nearest wastewater facility removed due to data availability concerns. We were unable to find a national dataset of wastewater treatment plant locations with permitted flow data, that was based on data generated within 5 years of other input datasets and included all relevant facilities. ZTRAX data were also removed due to data availability and replaced with a parcel value feature from Regrid. The accuracy of FS3 was acceptable and thus used for the Stage 2 Florida 'statewide' model.

Table S3: Variable Importance Scores for the Stage 2 Model

Rank	Importance	Variable	Description	Dataset
1	41201.38	TBAwGC250	Total Building Area within an intersecting 250m x 250m grid cell	Building Footprints
2	38587.03	Parc_Area	Parcel Area	Parcels
3	32643.58	NBApPA	Ratio of Non Building Area to Parcel Area	Building Footprints
4	30993.66	MPAIGC250	Median Parcel Area within an intersecting 250m x 250m grid cell	Parcels
5	30083.02	TBAwGC500	Total Building Area within an intersecting 500m x 500m grid cell	Building Footprints
6	22333.29	TBAwGC1000	Total Building Area within an intersecting 1000m x 1000m grid cell	Building Footprints
7	20945.39	MPAIGC1000	Median Parcel Area within an intersecting 1000m x 1000m grid cell	Parcels
8	16435.49	PIGC250	Number of Parcels within an intersecting 250m x 250m grid cell	Parcels
9	13976.57	PIGC500	Number of Parcels within an intersecting 500m x 500m grid cell	Parcels
10	13629.74	WWTP_Dist	Distance to the nearest Wastewater Treatment Facility	Wastewater Treatment Plants
11	10825.80	Incorp	Incorporated Place in US Census (Yes or No)	US Census Places
12	10296.74	MBAwGC250	Median Building Area within an intersecting 250m x 250m grid cell	Building Footprints
13	9047.93	PIGC1000	Number of Parcels within an intersecting 1000m x 1000m grid cell	Parcels
14	8639.97	B25037e2	Census Block MEDIAN YEAR STRUCTURE BUILT BY TENURE: Total: Owner occupied: Occupied housing units -- (Estimate)	US Census ACS
15	8278.03	MBAwGC500	Median Building Area within an intersecting 500m x 500m grid cell	Building Footprints
16	8030.62	parval_Fill	Parcel (Monetary) Value	Parcel Value
17	7883.79	PCBGC500	Number of Parcels completely contained within an intersecting 500m x 500m grid cell	Parcels
18	7818.52	B25037e1	Census Block MEDIAN YEAR STRUCTURE BUILT BY TENURE: Total: Occupied housing units -- (Estimate)	US Census ACS
19	7754.38	MBAwGC1000	Median Building Area within an intersecting 1000m x 1000m grid cell	Building Footprints
20	5622.32	B25010e1	Census Block AVERAGE HOUSEHOLD SIZE OF OCCUPIED HOUSING UNITS BY TENURE: Total: Occupied housing units -- (Estimate)	US Census ACS
21	5520.88	GC2000Num_S1400_max	Number of S1400 Roads (Local Neighborhood Roads, Rural Roads, City	Roads

			Streets) within an intersecting 2000m x 2000m grid cell	
22	5168.24	B25010e2	Census Block AVERAGE HOUSEHOLD SIZE OF OCCUPIED HOUSING UNITS BY TENURE: Total: Owner occupied: Occupied housing units -- (Estimate)	US Census ACS
23	5088.60	B25037e3	Census Block MEDIAN YEAR STRUCTURE BUILT BY TENURE: Total: Renter occupied: Occupied housing units -- (Estimate)	US Census ACS
24	5074.64	GC2000Len_S1400_max	Total length of S1400 Roads (Local Neighborhood Roads, Rural Roads, City Streets) within an intersecting 2000m x 2000m grid cell	Roads
25	4868.48	B25010e3	Census Block AVERAGE HOUSEHOLD SIZE OF OCCUPIED HOUSING UNITS BY TENURE: Total: Renter occupied: Occupied housing units -- (Estimate)	US Census ACS
26	4844.34	TotBldgArea	Total Building Area on Parcel	Building Footprints
27	4843.84	RdLenGC2000_max	Total length of roads within an intersecting 2000m x 2000m grid cell	Roads
28	4778.61	BCBGC1000	Number of Buildings completely contained within an intersecting 1000m x 1000m grid cell	Building Footprints
29	4695.03	B25041e4	Census Block BEDROOMS: Total: 2 bedrooms: Housing units -- (Estimate)	US Census ACS
30	4631.50	GC1000Len_S1400_max	Total length of S1400 Roads (Local Neighborhood Roads, Rural Roads, City Streets) within an intersecting 1000m x 1000m grid cell	Roads
31	4485.23	B25017e6	Census Block ROOMS: Total: 5 room: Housing units -- (Estimate)	US Census ACS
32	4395.10	B25041e6	Census Block BEDROOMS: Total: 4 bedrooms: Housing units -- (Estimate)	US Census ACS
33	4283.63	RdLenGC1000_max	Total length of roads within an intersecting 1000m x 1000m grid cell	Roads
34	4234.47	B25041e5	Census Block BEDROOMS: Total: 3 bedrooms: Housing units -- (Estimate)	US Census ACS
35	4188.79	BCBGC250	Number of Buildings completely contained within an intersecting 250m x 250m grid cell	Building Footprints
36	4096.18	B25017e7	Census Block ROOMS: Total: 6 room: Housing units -- (Estimate)	US Census ACS
37	4093.92	B25017e5	Census Block ROOMS: Total: 4 room: Housing units -- (Estimate)	US Census ACS
38	3964.63	GC1000Num_S1400_max	Total length of S1400 Roads (Local Neighborhood Roads, Rural Roads, City Streets) within an intersecting 1000m x 1000m grid cell	Roads

39	3957.91	B25017e1	Census Block ROOMS: Total: Housing units -- (Estimate)	US Census ACS
40	3794.80	B25017e8	Census Block ROOMS: Total: 7 room: Housing units -- (Estimate)	US Census ACS
41	3769.56	B25017e10	Census Block ROOMS: Total: 9 or more rooms: Housing units -- (Estimate)	US Census ACS
42	3765.24	BCBGC500	Number of Buildings completely contained within an intersecting 500m x 500m grid cell	Building Footprints
43	3627.55	B25017e9	Census Block ROOMS: Total: 8 room: Housing units -- (Estimate)	US Census ACS
44	3613.10	MaxBldgArea	Maximum Building Area on Parcel	Building Footprints
45	3556.22	GC2000Num_S1200_max	Number of S1200 Roads (Secondary Roads) within an intersecting 2000m x 2000m grid cell	Roads
46	3495.33	B25017e4	Census Block ROOMS: Total: 3 room: Housing units -- (Estimate)	US Census ACS
47	3094.96	Undesignated	Place Undesignated by US Census (Yes or No)	US Census Places
48	2640.01	B25041e3	Census Block BEDROOMS: Total: 1 bedroom: Housing units -- (Estimate)	US Census ACS
49	2560.58	B25041e7	Census Block BEDROOMS: Total: 5 or more bedrooms: Housing units -- (Estimate)	US Census ACS
50	2232.47	CDP	US Census Designated Place (Yes or No)	US Census Places
51	1921.04	NumBldgCont	Number of Buildings contained on the parcel	Building Footprints
52	1726.33	B25017e3	Census Block ROOMS: Total: 2 room: Housing units -- (Estimate)	US Census ACS
53	1626.25	GC2000Num_S1740_max	Number of S1740 Roads (Private Roads for service vehicles) within an intersecting 2000m x 2000m grid cell	Roads
54	1606.64	GC1000Num_S1200_max	Number of S1200 Roads (Secondary Roads) within an intersecting 1000m x 1000m grid cell	Roads
55	1332.29	B25041e2	Census Block BEDROOMS: Total: No bedroom: Housing units -- (Estimate)	US Census ACS
56	1250.08	B25017e2	Census Block ROOMS: Total: 1 room: Housing units -- (Estimate)	US Census ACS
57	817.62	GC2000Num_S1100_max	Number of S1100 Roads (Primary Roads) within an intersecting 2000m x 2000m grid cell	Roads
58	623.80	GC1000Num_S1740_max	Number of S1740 Roads (Private Roads for service vehicles) within an intersecting 1000m x 1000m grid cell	Roads
59	431.43	GC1000Num_S1100_max	Number of S1100 Roads (Primary Roads) within an intersecting 1000m x 1000m grid cell	Roads

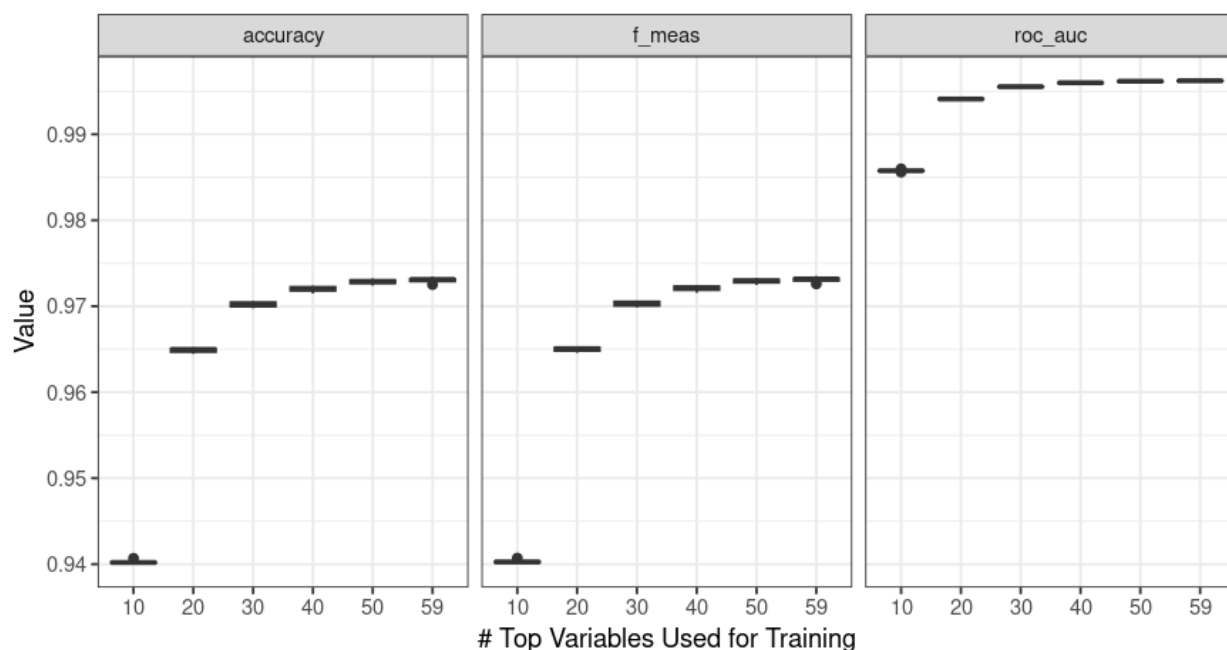


Figure S4: Model performance of vanilla random forest classifiers trained with default hyperparameters as a function of the number of top-ranked variables (10–59) included by variable importance score. Performance gains plateau after ~30 variables, suggesting minimal benefit from additional features.

Table S4: Percentage of predictions for ground truth versus predicted labels in each category relative to each category

		Prediction		
		NonApp	Onsite	Sewer
Ground Truth	NonApp	82.7%	7.3%	10.0%
	Onsite	2.1%	95.4%	2.5%
	Sewer	1.0%	2.9%	96.1%

Table S5: Confusion Matrix of Virginia ground truth labels versus predicted labels

		Prediction		
		NonApp	Onsite	Sewer
Ground Truth	NonApp	47172	8666	2647
	Onsite	18068	208614	5905
	Sewer	7989	24249	50660

Section S4: EMBRACE Checklist

We used the EMBRACE checklist (Zhu et al., 2024, 2023) to assure that the methods described in the paper followed best practices for machine learning in environmental research. The next pages include the checklist filled with the data relevant to this manuscript.

The EMBRACE Checklist

Version 1.0. Last Updated Date: September, 2024.

The main purpose of EMBRACE Checklist is to facilitate the reporting of basic data and methods used for supervised machine learning (ML) research in environmental science and engineering (ESE). It also aims to provide a comprehensive evaluation of important preparation and analysis steps. We warmly encourage researchers to include the checklist as a reference, and if interested list it as supplementary materials in their publications.

Detailed instructions and additional resources are available open access for download from the author's GitHub: <https://github.com/starfriend10/EMBRACE>

- Future updates and new resources will be available in the above GitHub repository, which is also served as a community-owned platform to share data and codes, refine forms/checklists, discuss ML-ESE topics, promote research outcomes, and continuously identify/avoid pitfalls and follow good practices.
- We encourage you to direct share your checklists, but you can also save it as a read-only document. See instructions for detailed steps.
- Please cite the Viewpoint when using the checklist. If you have any questions or suggestions, please contact Dr. Jun-Jie Zhu at Princeton University (junjiez@princeton.edu or ranmuweijie@gmail.com).

Project Overview			
Project Title	Look Out Below: Predicting Wastewater Infrastructure Service Type at the Land Parcel Scale		
Contributing Authors	Nelson da Luz, Jay Taneja, Emily Kumpel		
Date	06/24/2025	Completed by	Nelson da Luz
Contact Email	ndaluz@umass.edu	DOI (if published)	
Domain Category	None of above	Or Specify:	
Learning Type	Regression ☐	Classification ☒	Regression+ Classification ☐
Prediction Type	Deterministic ☒	Probabilistic ☐	Deterministic+ Probabilistic ☐
Other Information	This manuscript presents a series of models for predicting wastewater infrastructure service type. A baseline model is ☒		

I. Problem Formulation							
Project Objective(s)	The objective of this project is to develop methods for predicting whether a land parcel is served by sewer, an onsite wastewater treatment or disposal method, or neither. ☒						
Feasibility Assessment	Data Availability	Model Accessibility	Computation Resources	Knowledge Preparation	Time Availability	Financial Availability	Risk Tolerance
Levels	Medium	Medium	High	High	Medium	High	Medium

II. Data Collection								
Data Sources	☐	Time-series Monitoring	☐	Field Campaigns	☒	Government Database	☐	Scientific Literature
	☐	Laboratory Experiments	☐	Simulation Outputs	☒	Others. Specify:	Non-public gov't database	
Source Types	☒	Ordinary	☐	Time-series	☐	Literature-based	☐	Others. Specify:
Data Types	☒	Continuous Numerical	☒	Categorical	☒	Textual	☐	Visual
	☐	Graphical	☐	Auditory	☐	Others. Specify:		
Data Ethics	☒	No special ethical considerations are required, or followed the necessary requirements when needed						
	☒	No special permissions are required, or obtained the necessary permissions when needed.						
	☒	Provided appropriate credit(s) to the data source(s)						
Data Availability	☐	Data are publicly available through a PURL or DOI			☒	Data are available as requested		
Raw Data Quantity	Raw Sample Size		Raw Variable Size		Raw Total Data volume		Raw Sample-Variable Ratio	
	5,237,561		166				31,551.6	
Raw Data Quality	The raw data had an internal QA/QC before data retrieval?				Yes	☒	No	☐
	The raw data went through a peer-review or similar process?				Yes	☐	No	☒

III. Data Preprocessing									
Data Cleaning	☒	Abnormal Data Management	☐	Statistical Outlier Management	☐	Technical Outlier Management			
	☐	Data Correction due to Technical Limitations	☐	Others. Specify:					
Data Enrichment	☒	Missing Data Management	☐	Data Augmentation	☒	Data Aggregation	☒	Data Balancing	
	Upper sample ratio among groups:		5.14	☐	Others. Specify:				
Feature Engineering	☒	Initial Variable(s) Exclusion	☒	Feature Importance Analysis	☐	Feature Extraction			
	☒	Categorical Feature Transformation	☐	Dummy Feature Removal					
	☐	Ordinal Encoding	Encoded variable(s):						
	☒	Feature Selection	☐	Feature Scaling (for X)	☐	Feature Scaling (for Y)	☐	Circular Feature Scaling	
	☒	New Feature Development	☒	Ordinary Mathematical Transformation(s)	☐	Time-Dependent Feature Transformation(s)			
			☒	Spatial Feature Transformation(s)	☐	Others. Specify:			
	Sparse Dataset?	Yes ☐	No ☒	Preprocessed to increase information density?	Yes ☐	No ☒			
☒	Normality Test (before/after transformation)	☐	Others. Specify:						
Data Splitting	Framework	☐	Training-Testing	☐	(Pre-)Training-Validation-Testing	☒	Cross-Validation-Testing		
	Splitting Method	☒	Simple Random	☐	Grouped Random	☐	Simple Block	☒	Sub-Group R/B
	☒	Ensured training dataset covered most possible scenarios (e.g., different seasons)							
Final Data Quantity	Testing Ratio	Preprocessed Sample Size		Preprocessed Feature Size		Sample-Feature Ratio		Training SFR	
	0.080	1,428,571		59		24,213.0		16,949.0	

IV. Methods Selection and Model Initialization									
Methods Selection	☒	Selected supervised learning methods to be tested based on case needs (e.g., data, computation resources, etc.)?							
	Methods Studied: N ML and M Statistical				N =	4	M =	0	
	ML Family	☒	Tree-based	☒	Neural Network	☒	Others	SVM	
	ML Type	☒	Shallow ML	☐	Deep ML	Final Optimal Method(s): Random Forest			
Model Uncertainty	☐	Examined Random Seeds			☒	Replicated Modeling Process			
	☐	Assessed probabilistic prediction results			☐	Others. Specify:			

V. Model Evaluation and Model Optimization									
Model Evaluation	☒	Set a goal model performance before evaluation							
	Specify Evaluation Metrics: Accuracy, F-1 Score, ROC-AUC								
	☒	Studied under-fitting issue				☒	Studied over-fitting issue		
	☒	Final evaluation was based on the testing data				☐	Weighted metrics were used for imbalanced data		
	☐	ML significantly outperformed statistical models				☐	Statistical perform similar or even better than ML models		
Hyper-parameter Optimization	HPO Method	☐	No HPO	☐	Trial-and-Error/Experience	☒	Grid/Random Search (similar)	☐	Metaheuristics
	☐	Other HPO. Specify:							
	Number of methods tuned by HPO:				4	Number of hyperparameters searched:		2	
	Number of values in each hyperparameter:			☐	Continuous Search	☒	Discrete Search. Specify the number:		5
	CV Method	☐	No CV	☒	k-fold CV	☐	Leave-one-out CV	☐	Stratified CV
☐	Pre-defined or other CV. Specify:								

VI. Model Explanation				
Model Interpretability	☐	Reported model interpretability as intrinsic property	☒	Described tradeoff between interpretability and complexity
	☐	Explained limitations of low model interpretability	☐	Compared the methods with different interpretabilities
Model Explainability	☐	Reported the explanation method(s) used for FIA	Explanation method(s):	
	☒	Reminded high accuracy might not be trustable explanation	☒	Described application of explanation on the optimal model
	Reflected on understanding that explanations:		☒	couldn't cover all mechanisms
Causality	☒	Reflected on higher accuracy not implying logical causal relationships		
	☒	Reported understanding that explanation results do not indicate causality		
	☒	Described alignment of explanations with environmental domain knowledge for causality		
	☒	Identified illogical or counterintuitive parts based on environmental domain knowledge		
	☒	Described study of illogical or counterintuitive parts based on environmental domain knowledge		

VII. Data Leakage Management				
General/ Data Source	Verified no:	☒	response variable Y leakage	☐
	future-to-current data leakage		☐	current-to-current data leakage
	Confirmed that no overlapped data between training and testing		Yes	☒
Data splitting method for literature-source data:		No	☐	
Data Enrichment	Verified that missing data replacement utilized only seen dataset	N/A	☒	Yes ☐ No ☐
	Verified that missing data interpolation utilized neighbor data in same subset	N/A	☒	Yes ☐ No ☐
	Verified that model-based imputation utilized only seen data	N/A	☒	Yes ☐ No ☐
Feature Engineering	Confirmed that feature engineering utilized only seen dataset	N/A	☐	Yes ☒ No ☐
	Confirmed that feature scaling utilized only seen dataset	N/A	☐	Yes ☒ No ☐
	Confirmed that data splitting method for time-dependent data:			
CV Loop	Verified that missing data replacement utilized only seen data in each split	N/A	☒	Yes ☐ No ☐
	Verified that missing data interpolation with neighbor data in each split	N/A	☒	Yes ☐ No ☐
	Verified that feature engineering utilized only seen data in each split	N/A	☐	Yes ☒ No ☐
	Verified that feature scaling utilized only seen data in each split	N/A	☐	Yes ☒ No ☐
	Data splitting method for literature-source data in CV loop:			
	Data splitting method for time-dependent data in CV loop:			
Others	☒	Self-specify: Any missing data filling by replacement was performed at the county level so any effects related to data leakage		

VIII. Additional Items	
☐	PURL or DOI are publicly available. Data: Code:
☐	Email or URL are available to request. Data: Code:
☐	Self-specified Item:

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